Mathematical Foundations of Spline Geometry & Benchmark Estimands

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This document details the theoretical formulation of spline geometric metrics (Deviation-From-Linearity, DFL) and establishes the distinction between conditional and marginal estimands in KAN interpretability.

Proposition 1: Reference-Induced Deformation Decomposition

Let the empirical computation along an edge (i, j) in a Kolmogorov–Arnold Network be parameterized by learnable B-splines:

$$\Phi_{ij}(x) = W_{ij}^{base}\,\mathrm{SiLU}(x) + \sum_{k=1}^{K} w_{ij,k} B_k(x) \tag{1}$$

When evaluating the non-linearity of this function over a specified evaluation domain $X \subset \mathbb{R}$, we compute the Deviation-From-Linearity (DFL):

$$\mathrm{DFL}(\Phi_{ij}, X) = 1 - R^2 = \frac{\sum_{t\in X}(\Phi_{ij}(t) - \hat{y}(t))^2}{\sum_{t\in X}(\Phi_{ij}(t) - \bar{y})^2} \tag{2}$$

The Geometric Decomposition:
The measured geometric curvature over support X decomposes into two components:

$$G(f, X) = S(f) + D(X, f) \tag{3}$$

where $S(f)$ is the intrinsic mathematical transformation, and $D(X, f)$ is the domain-dependent deformation induced by evaluating over support X. As proven in Exp 2-B, a domain shift alone can inflate $D(X, f)$ by $+0.1195$ without altering the underlying function.

Proposition 2: Optimization Randomness and Estimand Duality

Because deep neural networks navigate non-convex loss landscapes, the learned spline parameters $w_{ij,k}(s)$ are stochastic realizations dependent on the random seed $s \sim \mathcal{S}$ (initialization and mini-batch sequence).

1. Conditional Estimand

For a fixed seed s, the conditional excess nonlinearity is defined as:

$$\Delta(s) = G(f_{\mathrm{real}}^{(s)}, X) - \mathbb{E}_{\pi}[G(f_{\mathrm{null}(\pi)}^{(s)}, X)] \tag{4}$$

Testing $H_0^{(s)} : \Delta(s) \leq 0$ evaluates whether real labels induced greater curvature *within the specific optimization basin reached by seed s*. For seed $s = 42$, $\Delta(42) = +0.0017$ ($p = 0.0060$).

2. Marginal Population Estimand

The scientific hypothesis of interest across the optimization distribution is the marginal expectation:

$$\Delta_{\mathrm{marginal}} = \mathbb{E}_{s\sim\mathcal{S}}[\Delta(s)] = \mathbb{E}_s[G(f_{\mathrm{real}}^{(s)}, X)] - \mathbb{E}_{s,\pi}[G(f_{\mathrm{null}}^{(s)}, X)] \tag{5}$$

Empirical Benchmark Finding:
Evaluating over 10 independent initialization seeds reveals:

$$\Delta_{\text{marginal}} = -0.00058 \leq 0 \quad (95\%\ \text{CI: } [-0.00120, +0.00003], \quad p = 0.9513) \tag{6}$$

Conclusion: Under the predefined one-sided alternative, the observed marginal effect does not support rejection of the null hypothesis. Single-seed conditional inference does not generalize across optimization randomness.